

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
MTH302	PROBABILITY AND STATISTICS	3	1	0	4	
Course Weightage	ATT: 5 CA: 25 MTT: 20 ETT: 50					
Course Focus	EMPLOYABILITY,SKILL DEVELOPMENT					

Course Outcomes :Through this course students should be able to

CO1 :: recall the concept of random variables and its distribution functions.

CO2 :: recognize relationships among the variables through correlation and regression.

CO3 :: apply probability distributions to find the solution of different engineering problems.

CO4 :: analyze estimators for unbiasedness, consistency, and efficiency, including maximum likelihood estimators

CO5 :: interpret the results of statistical tests such as Z-test, t-test, F-test, and Chi-square test for hypothesis testing

	TextBooks (T)		
Sr No	Title	Author	Publisher Name
T-1	PROBABILITY AND STATISTICS FOR ENGINEERS AND SCIENTISTS	RONALD E. WALPOLE, RAYMOND H. MYERS,SHARON L. MYERS, AND KEYING YE	PEARSON
	Reference Books (R)		
Sr No	Title	Author	Publisher Name
R-1	PROBABILITY AND STATISTICS FOR ENGINEERS	DR. J. RAVICHANDRAN	WILEY
R-2	FUNDAMENTALS OF MATHEMATICAL STATISTICS	S.C. GUPTA AND V.K. KAPOOR	SULTAN CHAND & SONS (P) LTD.

Relevant Websites (RW)		
Sr No	(Web address) (only if relevant to the course)	Salient Features
RW-1	https://byjus.com/maths/random-variables-and-its-probability-distributions/	Random Variables and Probability Distributions
RW-2	https://www.cuemath.com/data/correlation-and-regression/	Correlation and Linear regression
RW-3	https://www.probabilitycourse.com/chapter3/3_1_5_special_discrete_distr.php	Special Discrete Distributions
RW-4	https://www.educative.io/answers/continuous-probability-distribution	Special Continuous Distributions

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

RW-5	https://corporatefinanceinstitute.com/resources/data-science/point-estimators/	Point Estimation
RW-6	https://byjus.com/jee/central-limit-theorem/	Central Limit Theorem
RW-7	https://www.scribbr.com/statistics/hypothesis-testing/	Testing of Hypothesis

Audio Visual Aids (AV)		
Sr No	(AV aids) (only if relevant to the course)	Salient Features
AV-1	https://www.youtube.com/watch?v=zujeSyREcQ4	Random variables and its Characterization
AV-2	https://www.youtube.com/watch?v=WMUMc2QJrPQ	Correlation and Regression
AV-3	https://www.youtube.com/watch?v=6x1pL9Yov1k	Discrete probability distributions
AV-4	https://www.youtube.com/watch?v=wcexipNgbhs	Special Continuous Distributions
AV-5	https://www.youtube.com/watch?v=pu2OzGhv1HI	Point Estimation
AV-6	https://www.youtube.com/watch?v=-FtlH4svqx4	Hypothesis Testing
AV-7	https://www.youtube.com/watch?v=ugidzgLMZyA	Spearman's Rank correlation coefficient
AV-8	https://www.youtube.com/watch?v=PjihjKfBuzc	exponential distribution and its properties
AV-9	https://www.youtube.com/watch?v=2QeDRsxSF9M&t=449s	chi square test for goodness of fit
AV-10	https://www.youtube.com/watch?v=_jZ7oL2rkKA	Mean and variance of random variables
AV-11	https://www.youtube.com/watch?v=sHbX58y5D4U	Scatter plots
AV-12	https://www.youtube.com/watch?v=0QbjCAK4ijw	Maximum likelihood estimation
AV-13	https://www.youtube.com/watch?v=Hdbbx7DIweQ	Types of error
AV-14	https://www.youtube.com/watch?v=zJ8e_wAWUzE	Z-test and Student's t-test
AV-15	https://www.youtube.com/watch?v=vWWwaoK_Qaw	F-test

LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	7

Detailed Plan For Lectures

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples

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Week 1	Lecture 1	Random Variables and Probability Distributions(discrete and continuous random variables and their probability distributions)	T-1	RW-1 AV-1	In Lecture 1, zero lecture will be covered along with basic details of random variables. In Lecture 2, discrete and continuous random variables and their distribution functions will be discussed.	understand discrete and continuous random variables and their distribution functions	Discussion along with ppt and white board.	Fast Food Survey Which fast food option do you prefer? 1. Domino's Pizza 2. McDonald's Burger 3. Subway Salad We'll use the responses of the student to introduce random variables and probability distributions
	Lecture 2	Random Variables and Probability Distributions(discrete and continuous random variables and their probability distributions)	T-1	RW-1 AV-1	In Lecture 1, zero lecture will be covered along with basic details of random variables. In Lecture 2, discrete and continuous random variables and their distribution functions will be discussed.	understand discrete and continuous random variables and their distribution functions	Discussion along with ppt and white board.	Fast Food Survey Which fast food option do you prefer? 1. Domino's Pizza 2. McDonald's Burger 3. Subway Salad We'll use the responses of the student to introduce random variables and probability distributions
	Lecture 3	Random Variables and Probability Distributions(cumulative distribution functions)	T-1	RW-1 AV-1	Introduction to the definition, properties, and applications of cumulative distribution functions for discrete and continuous random variables.	Students will be able to define, interpret, and apply cumulative distribution functions to compute probabilities for discrete and continuous random variables.	Discussion along with ppt and white board.	The CDF can be used to find the probability that a randomly chosen person's height is less than or equal to 170 cm in a population.

Week 2	Lecture 4	Random Variables and Probability Distributions(moments about origin and mean (central moments) up to fourth order)	R-2	RW-1 AV-10	Introduction to moments about origin and mean (up to fourth order) in probability distributions to measure mean, variance, skewness, and kurtosis for understanding distribution shape and spread.	Students will be able to calculate and interpret moments of a probability distribution to analyze its central tendency, variability, asymmetry, and peakedness.	Discussion along with ppt and white board.	Collect the marks of students from a recent test, calculate the mean, variance, skewness, and kurtosis of the score distribution, and discuss whether the distribution is symmetric, skewed, or has heavy/light tails.
		Random Variables and Probability Distributions(mean)	R-2	RW-1 AV-10	Introduction to moments about origin and mean (up to fourth order) in probability distributions to measure mean, variance, skewness, and kurtosis for understanding distribution shape and spread.	Students will be able to calculate and interpret moments of a probability distribution to analyze its central tendency, variability, asymmetry, and peakedness.	Discussion along with ppt and white board.	Collect the marks of students from a recent test, calculate the mean, variance, skewness, and kurtosis of the score distribution, and discuss whether the distribution is symmetric, skewed, or has heavy/light tails.
		Random Variables and Probability Distributions(variance)	R-2	RW-1 AV-10	Introduction to moments about origin and mean (up to fourth order) in probability distributions to measure mean, variance, skewness, and kurtosis for understanding distribution shape and spread.	Students will be able to calculate and interpret moments of a probability distribution to analyze its central tendency, variability, asymmetry, and peakedness.	Discussion along with ppt and white board.	Collect the marks of students from a recent test, calculate the mean, variance, skewness, and kurtosis of the score distribution, and discuss whether the distribution is symmetric, skewed, or has heavy/light tails.

Week 2	Lecture 4	Random Variables and Probability Distributions(skewness)	R-2	RW-1 AV-10	Introduction to moments about origin and mean (up to fourth order) in probability distributions to measure mean, variance, skewness, and kurtosis for understanding distribution shape and spread.	Students will be able to calculate and interpret moments of a probability distribution to analyze its central tendency, variability, asymmetry, and peakedness.	Discussion along with ppt and white board.	Collect the marks of students from a recent test, calculate the mean, variance, skewness, and kurtosis of the score distribution, and discuss whether the distribution is symmetric, skewed, or has heavy/light tails.
		Random Variables and Probability Distributions(kurtosis)	R-2	RW-1 AV-10	Introduction to moments about origin and mean (up to fourth order) in probability distributions to measure mean, variance, skewness, and kurtosis for understanding distribution shape and spread.	Students will be able to calculate and interpret moments of a probability distribution to analyze its central tendency, variability, asymmetry, and peakedness.	Discussion along with ppt and white board.	Collect the marks of students from a recent test, calculate the mean, variance, skewness, and kurtosis of the score distribution, and discuss whether the distribution is symmetric, skewed, or has heavy/light tails.
	Lecture 5	Random Variables and Probability Distributions(moments about origin and mean (central moments) up to fourth order)	R-2	RW-1 AV-10	Introduction to moments about origin and mean (up to fourth order) in probability distributions to measure mean, variance, skewness, and kurtosis for understanding distribution shape and spread.	Students will be able to calculate and interpret moments of a probability distribution to analyze its central tendency, variability, asymmetry, and peakedness.	Discussion along with ppt and white board.	Collect the marks of students from a recent test, calculate the mean, variance, skewness, and kurtosis of the score distribution, and discuss whether the distribution is symmetric, skewed, or has heavy/light tails.

Week 2	Lecture 5	Random Variables and Probability Distributions(mean)	R-2	RW-1 AV-10	Introduction to moments about origin and mean (up to fourth order) in probability distributions to measure mean, variance, skewness, and kurtosis for understanding distribution shape and spread.	Students will be able to calculate and interpret moments of a probability distribution to analyze its central tendency, variability, asymmetry, and peakedness.	Discussion along with ppt and white board.	Collect the marks of students from a recent test, calculate the mean, variance, skewness, and kurtosis of the score distribution, and discuss whether the distribution is symmetric, skewed, or has heavy/light tails.
		Random Variables and Probability Distributions(variance)	R-2	RW-1 AV-10	Introduction to moments about origin and mean (up to fourth order) in probability distributions to measure mean, variance, skewness, and kurtosis for understanding distribution shape and spread.	Students will be able to calculate and interpret moments of a probability distribution to analyze its central tendency, variability, asymmetry, and peakedness.	Discussion along with ppt and white board.	Collect the marks of students from a recent test, calculate the mean, variance, skewness, and kurtosis of the score distribution, and discuss whether the distribution is symmetric, skewed, or has heavy/light tails.
		Random Variables and Probability Distributions(skewness)	R-2	RW-1 AV-10	Introduction to moments about origin and mean (up to fourth order) in probability distributions to measure mean, variance, skewness, and kurtosis for understanding distribution shape and spread.	Students will be able to calculate and interpret moments of a probability distribution to analyze its central tendency, variability, asymmetry, and peakedness.	Discussion along with ppt and white board.	Collect the marks of students from a recent test, calculate the mean, variance, skewness, and kurtosis of the score distribution, and discuss whether the distribution is symmetric, skewed, or has heavy/light tails.

Week 2	Lecture 5	Random Variables and Probability Distributions(kurtosis)	R-2	RW-1 AV-10	Introduction to moments about origin and mean (up to fourth order) in probability distributions to measure mean, variance, skewness, and kurtosis for understanding distribution shape and spread.	Students will be able to calculate and interpret moments of a probability distribution to analyze its central tendency, variability, asymmetry, and peakedness.	Discussion along with ppt and white board.	Collect the marks of students from a recent test, calculate the mean, variance, skewness, and kurtosis of the score distribution, and discuss whether the distribution is symmetric, skewed, or has heavy/light tails.
	Lecture 6	Random Variables and Probability Distributions(moments about origin and mean (central moments) up to fourth order)	R-2	RW-1 AV-10	Introduction to moments about origin and mean (up to fourth order) in probability distributions to measure mean, variance, skewness, and kurtosis for understanding distribution shape and spread.	Students will be able to calculate and interpret moments of a probability distribution to analyze its central tendency, variability, asymmetry, and peakedness.	Discussion along with ppt and white board.	Collect the marks of students from a recent test, calculate the mean, variance, skewness, and kurtosis of the score distribution, and discuss whether the distribution is symmetric, skewed, or has heavy/light tails.
		Random Variables and Probability Distributions(mean)	R-2	RW-1 AV-10	Introduction to moments about origin and mean (up to fourth order) in probability distributions to measure mean, variance, skewness, and kurtosis for understanding distribution shape and spread.	Students will be able to calculate and interpret moments of a probability distribution to analyze its central tendency, variability, asymmetry, and peakedness.	Discussion along with ppt and white board.	Collect the marks of students from a recent test, calculate the mean, variance, skewness, and kurtosis of the score distribution, and discuss whether the distribution is symmetric, skewed, or has heavy/light tails.

Week 2	Lecture 6	Random Variables and Probability Distributions(variance)	R-2	RW-1 AV-10	Introduction to moments about origin and mean (up to fourth order) in probability distributions to measure mean, variance, skewness, and kurtosis for understanding distribution shape and spread.	Students will be able to calculate and interpret moments of a probability distribution to analyze its central tendency, variability, asymmetry, and peakedness.	Discussion along with ppt and white board.	Collect the marks of students from a recent test, calculate the mean, variance, skewness, and kurtosis of the score distribution, and discuss whether the distribution is symmetric, skewed, or has heavy/light tails.
		Random Variables and Probability Distributions(skewness)	R-2	RW-1 AV-10	Introduction to moments about origin and mean (up to fourth order) in probability distributions to measure mean, variance, skewness, and kurtosis for understanding distribution shape and spread.	Students will be able to calculate and interpret moments of a probability distribution to analyze its central tendency, variability, asymmetry, and peakedness.	Discussion along with ppt and white board.	Collect the marks of students from a recent test, calculate the mean, variance, skewness, and kurtosis of the score distribution, and discuss whether the distribution is symmetric, skewed, or has heavy/light tails.
		Random Variables and Probability Distributions(kurtosis)	R-2	RW-1 AV-10	Introduction to moments about origin and mean (up to fourth order) in probability distributions to measure mean, variance, skewness, and kurtosis for understanding distribution shape and spread.	Students will be able to calculate and interpret moments of a probability distribution to analyze its central tendency, variability, asymmetry, and peakedness.	Discussion along with ppt and white board.	Collect the marks of students from a recent test, calculate the mean, variance, skewness, and kurtosis of the score distribution, and discuss whether the distribution is symmetric, skewed, or has heavy/light tails.

Week 3	Lecture 7	Correlation and Linear regression(scatter plots)	R-1 R-2	RW-2 AV-11	Scatter plots and Coefficient of Correlation and its properties	Learn about the scatter plots to find the kind of correlation between two random variables.	Discussion along with white board work and problem solving	To understand different scatter plots
		Correlation and Linear regression(correlation coefficient and its properties)	R-1 R-2	RW-2 AV-11	Scatter plots and Coefficient of Correlation and its properties	Learn about the scatter plots to find the kind of correlation between two random variables.	Discussion along with white board work and problem solving	To generate correlation matrix heatmap
	Lecture 8	Correlation and Linear regression(Karl Pearson's correlation coefficient)	R-1 R-2	RW-2 AV-2	Karl Pearson's correlation coefficient	learn methods to find correlation coefficient given by Karl Pearson	Flipped Classroom	A teacher wants to analyze whether there is a linear relationship between the number of hours a student studies and their exam scores.
	Lecture 9	Correlation and Linear regression(Spearman's rank correlation coefficient)	R-2	AV-7	Spearman's rank correlation coefficient	To learn the basics of Spearman's rank correlation coefficient	Discussion along with white board work and problem solving	
Week 4	Lecture 10	Correlation and Linear regression(Spearman's rank correlation coefficient)	R-2	AV-7	Spearman's rank correlation coefficient	To learn the basics of Spearman's rank correlation coefficient	Discussion along with white board work and problem solving	

Week 4	Lecture 11	Correlation and Linear regression(Linear regression and its properties)	R-1 R-2	RW-2 AV-2	Linear regression and its properties	understand Linear Regression and Properties of Regression Coefficients	Experiential Learning (Hands-On Activities): Dataset Analysis: Provide datasets (e.g., income vs. expenditure). Use Excel as a tool to calculate regression coefficients. Live Demonstrations: Demonstrate step-by-step fitting of regression lines and calculation of coefficients. Show how to interpret slope and intercept values.	
	Lecture 12	Correlation and Linear regression(Linear regression and its properties)	R-1 R-2	RW-2 AV-2	Linear regression and its properties	understand Linear Regression and Properties of Regression Coefficients	Experiential Learning (Hands-On Activities): Dataset Analysis: Provide datasets (e.g., income vs. expenditure). Use Excel as a tool to calculate regression coefficients. Live Demonstrations: Demonstrate step-by-step fitting of regression lines and calculation of coefficients. Show how to interpret slope and intercept values.	
Week 5	Lecture 13	Special Discrete Distributions(bernoulli process)	T-1	RW-3 AV-3	the Bernoulli process and Binomial distribution	learn the Bernoulli process and binomial distribution		

Week 5	Lecture 13	Special Discrete Distributions(binomial distribution)	T-1	RW-3 AV-3	the Bernoulli process and Binomial distribution	learn the Bernoulli process and binomial distribution	Discussion along with white board work.	A new vaccine has a 70% success rate. In a trial with 20 patients, what is the probability that exactly 15 patients will respond positively to the vaccine?
	Lecture 14	Special Discrete Distributions(bernoulli process)	T-1	RW-3 AV-3	the Bernoulli process and Binomial distribution	learn the Bernoulli process and binomial distribution		
		Special Discrete Distributions(binomial distribution)	T-1	RW-3 AV-3	the Bernoulli process and Binomial distribution	learn the Bernoulli process and binomial distribution	Discussion along with white board work.	A new vaccine has a 70% success rate. In a trial with 20 patients, what is the probability that exactly 15 patients will respond positively to the vaccine?
	Lecture 15	Special Discrete Distributions(negative binomial distribution)	T-1	RW-3 AV-3	negative binomial distribution and geometric distribution	learn negative binomial distribution and geometric distribution of a random variable	Discussion along with white board work.	
		Special Discrete Distributions(geometric distribution)	T-1	RW-3 AV-3	negative binomial distribution and geometric distribution	learn negative binomial distribution and geometric distribution of a random variable	Discussion along with white board work.	
Week 6	Lecture 16				Test 1			
	Lecture 17	Special Discrete Distributions(poisson distribution)	T-1	RW-3 AV-3	Poisson distribution	learn the Poisson distribution and the Poisson process	Discussion along with white board work.	
	Lecture 18	Special Discrete Distributions(poisson distribution)	T-1	RW-3 AV-3	Poisson distribution	learn the Poisson distribution and the Poisson process	Discussion along with white board work.	

Week 7	Lecture 19	Special Discrete Distributions(moment generating function (without proof) of the above mentioned distributions.)	R-2	RW-3 AV-3	moment generating function (without proof) of bernoulli, binomial, negative binomial, geometric and poisson distribution	learn about mgf of the discrete probability distributions and its use in problem solving	Discussion along with white board work.	
		SPILL OVER						
Week 7	Lecture 20				Spill Over			
	Lecture 21				Spill Over			
		MID-TERM						
Week 8	Lecture 22	Special Continuous Distributions(exponential distribution)	T-1	RW-4 AV-8	Introduction to exponential and gamma distributions, their properties, parameters, and applications in modeling waiting times and lifetimes of events.	Students will be able to identify and apply exponential and gamma distributions to solve problems involving time-to-event data.	Discussion along with ppt and white board.	Measure and record the time gap between arrivals of students at the classroom door, then model the data using an exponential distribution and extend to a gamma distribution for multiple arrivals.
		Special Continuous Distributions(gamma distribution)	T-1	RW-4 AV-4	Introduction to exponential and gamma distributions, their properties, parameters, and applications in modeling waiting times and lifetimes of events.	Students will be able to identify and apply exponential and gamma distributions to solve problems involving time-to-event data.	Discussion along with ppt and white board.	Measure and record the time gap between arrivals of students at the classroom door, then model the data using an exponential distribution and extend to a gamma distribution for multiple arrivals.

Week 8	Lecture 23	Special Continuous Distributions(normal distribution)	T-1	RW-4 AV-4	Introduction to the normal distribution, its properties, and applications, along with the use of normal approximation to the binomial distribution for large sample sizes.	Students will be able to analyze problems using the normal distribution and apply normal approximation to the binomial to simplify probability calculations.	Discussion along with ppt and white board.	Measure the heights of students in the class, show how they roughly follow a normal distribution, and then use that distribution to estimate the probability of a student being taller than a given height.
		Special Continuous Distributions(normal approximation to the binomial)	T-1	RW-4 AV-4	Introduction to the normal distribution, its properties, and applications, along with the use of normal approximation to the binomial distribution for large sample sizes.	Students will be able to analyze problems using the normal distribution and apply normal approximation to the binomial to simplify probability calculations.	Discussion along with ppt and white board.	Measure the heights of students in the class, show how they roughly follow a normal distribution, and then use that distribution to estimate the probability of a student being taller than a given height.
	Lecture 24	Special Continuous Distributions(normal distribution)	T-1	RW-4 AV-4	Introduction to the normal distribution, its properties, and applications, along with the use of normal approximation to the binomial distribution for large sample sizes.	Students will be able to analyze problems using the normal distribution and apply normal approximation to the binomial to simplify probability calculations.	Discussion along with ppt and white board.	Measure the heights of students in the class, show how they roughly follow a normal distribution, and then use that distribution to estimate the probability of a student being taller than a given height.

Week 8	Lecture 24	Special Continuous Distributions(normal approximation to the binomial)	T-1	RW-4 AV-4	Introduction to the normal distribution, its properties, and applications, along with the use of normal approximation to the binomial distribution for large sample sizes.	Students will be able to analyze problems using the normal distribution and apply normal approximation to the binomial to simplify probability calculations.	Discussion along with ppt and white board.	Measure the heights of students in the class, show how they roughly follow a normal distribution, and then use that distribution to estimate the probability of a student being taller than a given height.
Week 9	Lecture 25	Special Continuous Distributions(central limit theorem (without proof))	T-1	RW-6	Introduction to the Central Limit Theorem (CLT) explaining how, for a large sample size, the sampling distribution of the sample mean approaches a normal distribution regardless of the original population distribution.	Students will be able to apply the CLT to approximate probabilities and make inferences about population parameters from sample means.	Discussion along with ppt and white board.	Ask each student to roll a die 30 times, record the mean of their rolls, and plot all sample means to visually show how they form an approximate normal distribution even though a single die roll is not normally distributed.
	Lecture 26	Special Continuous Distributions(moment generating function (without proof) of the above mentioned distributions.)	T-1	RW-4 AV-4	Discussion on the definition and use of the moment generating function (MGF) to derive moments of a distribution, focusing on the exponential, gamma, and normal distributions.	Students will be able to compute the MGF of exponential, gamma, and normal distributions, and use it to obtain mean, variance, and higher-order moments.	Discussion along with ppt and white board.	Model the lifetime of light bulbs using an exponential distribution, rainfall accumulation with a gamma distribution, and exam scores with a normal distribution, then use MGFs to calculate their means and variances.

Week 9	Lecture 27	Point Estimation(unbiased estimator)	R-2	RW-5 AV-5	Introduction to point estimation as the process of providing a single best estimate of an unknown population parameter, and the concept of unbiased estimators whose expected value equals the true parameter value.	Students will be able to define point estimation, identify unbiased estimators, and evaluate their performance in estimating population parameters.	Discussion along with ppt and white board.	Estimate the average height of all students in the college by taking the mean height from a random sample, and discuss whether the sample mean is an unbiased estimator of the population mean.
Week 10	Lecture 28	Point Estimation(unbiased estimator)	R-2	RW-5 AV-5	Introduction to point estimation as the process of providing a single best estimate of an unknown population parameter, and the concept of unbiased estimators whose expected value equals the true parameter value.	Students will be able to define point estimation, identify unbiased estimators, and evaluate their performance in estimating population parameters.	Discussion along with ppt and white board.	Estimate the average height of all students in the college by taking the mean height from a random sample, and discuss whether the sample mean is an unbiased estimator of the population mean.
	Lecture 29	Point Estimation(consistent estimator)	R-2	RW-5	Introduction to consistent estimators, which are estimators that converge in probability to the true value of a parameter as the sample size increases.	Students will be able to define consistency, verify whether an estimator is consistent, and understand its importance in large-sample estimation.	Discussion along with ppt and white board.	Take random samples of different sizes from students' test scores in the class, compute the sample mean each time, and show how larger samples produce estimates closer to the actual class average.
	Lecture 30	Point Estimation(efficient estimator)	R-2	RW-5 AV-5	Introduction to efficient estimators, which are unbiased estimators having the smallest variance among all consistent estimators for a parameter.	Students will be able to identify efficient estimators and distinguish between finite-sample and asymptotic efficiency.	Discussion along with ppt and white board	Compare sample mean and weighted mean of students' heights to show lower variance and consistency in large samples.

Week 11	Lecture 31	Point Estimation(maximum likelihood estimation)	R-2	AV-12	Introduction to the Maximum Likelihood Estimator (MLE) as a method of estimating parameters by maximizing the likelihood function based on observed data.	Students will be able to construct likelihood functions, find parameter values that maximize them, and interpret MLE properties such as consistency and asymptotic efficiency.	Discussion along with ppt and white board.	Use a sample of lifetimes of light bulbs to estimate the failure rate parameter of an exponential distribution by maximizing the likelihood function.
	Lecture 32	Point Estimation(maximum likelihood estimation)	R-2	AV-12	Introduction to the Maximum Likelihood Estimator (MLE) as a method of estimating parameters by maximizing the likelihood function based on observed data.	Students will be able to construct likelihood functions, find parameter values that maximize them, and interpret MLE properties such as consistency and asymptotic efficiency.	Discussion along with ppt and white board.	Use a sample of lifetimes of light bulbs to estimate the failure rate parameter of an exponential distribution by maximizing the likelihood function.
	Lecture 33	Testing of Hypothesis(types of error)	R-1 R-2	AV-13	types of Error	learn the types of error in sampling theory	Discussion along with white board work and problem solving.	
Week 12	Lecture 34				Test 2			
	Lecture 35	Testing of Hypothesis(Z-test for single mean and difference of means)	R-2	RW-7 AV-6 AV-14	Z-test for single mean and difference of means	To learn Z-test for single mean and difference of means	Discussion along with white board work and problem solving.	
	Lecture 36	Testing of Hypothesis (Student's t-test for single mean and difference of means)	R-2	RW-7 AV-6 AV-14	Student's t-test for single mean and difference of means	To learn Student's t-test for single mean and difference of means	Flipped classroom approach (In-class activity)	
Week 13	Lecture 37	Testing of Hypothesis (Student's t-test for single mean and difference of means)	R-2	RW-7 AV-6 AV-14	Student's t-test for single mean and difference of means	To learn Student's t-test for single mean and difference of means	Flipped classroom approach (In-class activity)	
	Lecture 38	Testing of Hypothesis(F-test)	R-2	RW-7 AV-6 AV-15	F-test	learn about F-test in sampling distribution	Discussion along with white board work and problem solving	

Week 13	Lecture 39	Testing of Hypothesis(Chi-square test for goodness of fit)	R-2	RW-7 AV-6 AV-9	Chi-square test for goodness of fit	to learn the Chi square test for goodness of fit	TExperiential learning through real-time classroom data collection and Chi-square analysis.	Collect data from students on preferred study time (morning/evening) and exam format (MCQ/descriptive), then perform a Chi-square test to check for association.
Week 14	Lecture 40	Testing of Hypothesis(Chi-square test for goodness of fit)	R-2	RW-7 AV-6 AV-9	Chi-square test for goodness of fit	to learn the Chi square test for goodness of fit	TExperiential learning through real-time classroom data collection and Chi-square analysis.	Collect data from students on preferred study time (morning/evening) and exam format (MCQ/descriptive), then perform a Chi-square test to check for association.
		SPILL OVER						
Week 14	Lecture 41				Spill Over			
	Lecture 42				Spill Over			
Week 15	Lecture 43				Spill Over			
	Lecture 44				Spill Over			
	Lecture 45				Spill Over			

Scheme for CA:

CA Category of this Course Code is:A0202 (2 out of 2)

Component	Weightage (%)	Mapped CO(s)
Recoup CA		
Test 2	50	
Test 1	50	

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allotment / submission Week
Test 2	To assess students' understanding of the exponential, gamma, and normal distributions; the normal approximation to the binomial and the CLT (statements only); MGFs (statements only); and core estimati	Based on topic exponential distribution, gamma distribution, normal distribution, normal approximation to the binomial, central limit theorem (without proof), moment generating function (without proof) of the above mentioned distributions, unbiased estimator, consistent estimator, efficient estimator, maximum likelihood estimation	Individual	Offline	30	11 / 12
Test 1	To evaluate students' understanding and application of random variables/distributions (pmf, pdf, CDF, moments to fourth order), simple linear regression (plots, fit, assumptions), and correlation (Pea	Based on topic discrete and continuous random variables and their probability distributions, cumulative distribution functions, moments about origin and mean (central moments) up to fourth order, mean, variance, skewness, kurtosis, scatter plots, correlation coefficient and its properties, Karl Pearson's correlation coefficient, Spearman's rank correlation coefficient, Linear regression and its properties	Individual	Offline	30	5 / 6
Recoup CA	subjective test based on the syllabus	Additional chance of CA	Individual	Online	30	8 / 10

Plan for Tutorial: (Please do not use these time slots for syllabus coverage)

Tutorial No.	Lecture Topic	Type of pedagogical tool(s) planned (case analysis,problem solving test,role play,business game etc)
Tutorial1	Solving Numerical Problems on Random Variables and Their Probability Distributions	Problem Solving
Tutorial2	Problem-Based Learning: Moments and Shape Characteristics of Distributions	Problem Solving

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Tutorial3	Worked Examples and Practice Problems: Scatter Plots and Karl Pearson's Correlation Coefficient	Problem Solving
Tutorial4	Problem-Based Learning: Rank Correlation, Regression Equations, and Their Properties	Problem Solving
Tutorial5	Problem Solving on Bernoulli Process, Binomial and Negative Binomial Distributions	Problem Solving
Tutorial6	Problem Solving on Geometric and Poisson Distributions with Moment Generating Functions	Problem Solving
Tutorial7	Pre-MTE Practice: MCQ Problems on Discrete Distributions and MGFs	Problem Solving
After Mid-Term		
Tutorial8	Problem Solving on Exponential and Gamma Distributions	Problem Solving
Tutorial9	Problem Solving on Normal Distribution and Normal Approximation to the Binomial	Problem Solving
Tutorial10	Problem Solving on Central Limit Theorem and Moment Generating Functions (Without Proof)	Problem Solving
Tutorial11	Problem Solving on Unbiased, Consistent, and Efficient Estimators	Problem Solving
Tutorial12	Problem Solving on Maximum Likelihood Estimation and types of error	Problem Solving
Tutorial13	Problem Solving on Z-test, and Student's t-test	Problem Solving
Tutorial14	Problem Solving on F-test and Chi-square Goodness of Fit Test	Problem Solving

